higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1	1.1.1	A		
	1.1.2	C		
	1.1.3	D		
	1.1.4	B		
	1.1.5	A		
			(5 × 1)	(5)
1.2	1.2.1	True		
	1.2.2	False		
	1.2.3	True		
	1.2.4	True		
	1.2.5	False		
			(5 × 1)	(5)
				[10]

QUESTION 2

2.1 2.1.1

- Overalls
- Safety boots
- Welding helmet
- Apron
- Leather spats
- Leather gloves
- Ear plugs
- Safety harness (Any 5 × 1) (5)

2.1.2

- An overall shields the body from flames, heat and burns.
- Safety boots protect the feet from falling objects, sparks and sharp debris.
- A welding helmet shields the eyes and face from sparks, heat and the brightness of the welding arc.
- An apron protects the overalls and the body from the heat generated during welding.
- Leather spats prevent spatter and slag from entering boots and protect the feet from burns.
- Leather gloves shelter the hands from sparks and the ultra-violet arc.
- Ear plugs protect the ears from loud noises.
- A safety harness prevents a person working at a height from falling. (Any 5 × 1) (5)

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2.2 Ensure that the necessary tools, welding equipment and PPE are available before starting the welding process.✓ Gather all relevant material that is to be used, and properly prepare it by removing any rust, paint and grease.✓ Create any bevels that are required to be done on the materials and clean to prevent any contamination of the weld.✓ Ensure that the correct electrodes are used and that there are no defects in the electrodes.✓ Make sure that there is power supply for the welding machine, and that the welding cables are secured. The correct amperage must be set according to the welding specification chart before welding starts.✓ The surrounding area must be cleaned before welding can start.✓ (6)

2.3 2.3.1
- Ensure that the correct electrodes are used.
- Ensure that the arc ray is not too bright.
- Ensure that the current is not too high.
- Ensure that the arc force is set correctly.
- Ensure that the material is cleaned properly. (Any 2 × 1)

2.3.2
- Ensure that the current is set correctly.
- Ensure that the electrode does not have any defects.
- Ensure that the current is neither too high nor too low.
- Ensure that the arc length is correct. The arc must not be too high or too low. (Any 2 × 1)

(2 × 2) (4)

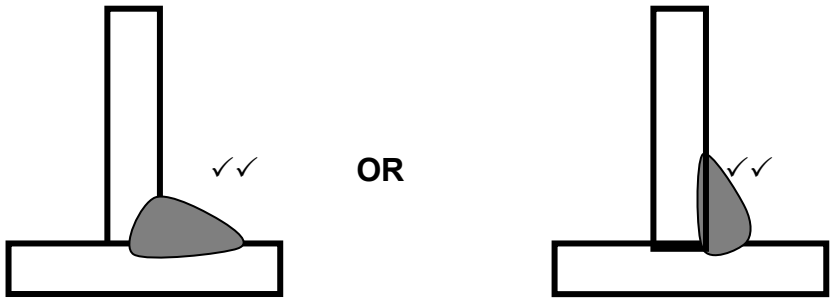
2.4
- Ensure that all chemical containers are sealed properly before storing.
- Ensure that there is adequate ventilation.
- Ensure that the chemical containers are cleaned properly after they have been used and that they are stored away on the correct shelves.
- Chemicals that have a reaction with other chemicals must be stored separately.
- Ensure that flammable chemicals are stored away from ignition sources.
- Ensure that the shelving is secured to prevent collapsing and spillage of chemicals. (Any 5 × 1) (5)

[25]

QUESTION 3

3.1
- Retrieve gas cylinders from the store, place them in an upright position and secure them. Open gas cylinder valves quickly and shut to remove any impurities on the valve.
- Connect gas regulators to gas cylinder valves according to their colours and type of thread, and turn regulator knobs to zero.
- Connect flashback arresters to regulators.
- Connect gas hoses to flashback arrestors. Open gas cylinder valves slowly and turn regulator knobs slowly to remove any impurities from gas hoses. Close cylinder valves and release regulator knobs.
- Connect flashback arrestors and gas welding torches to gas hoses. Open oxy-acetylene system and turn regulator knobs slowly and then check for any leaks. (5)

3.2



UNEQUAL WELD ON FILLET JOINT

(2)

3.3 A – Hammer or ball peen hammer
B – Work piece or welding joint
C – Vice (3)

[10]

QUESTION 4

4.1 To cut mild carbon steel, make sure that the oxy-acetylene flame on the cutting torch is set to a neutral flame.✓ Hold the cutting torch perpendicular to the work-piece that needs to be cut.✓ At the starting point, hold the flame slightly above the line on the work piece.✓ Heat the work piece until you see a bright red spot.✓ Press the cutting lever slowly until the oxygen flame has pierced the metal and sparks occur beneath the work piece.✓ Hold the cutting lever down and move the torch forward at the required speed until the work-piece is cut completely.✓ Don't move the cutting torch too slowly; this will cause the metal to burn away and slag to build up.✓ (7)

4.2 4.2.1 Gas cylinders are made of steel plate and are used to store gas. (1)

4.2.2 A content pressure gauge is used for checking the pressure in the cylinder. (2)

4.2.3 Gas hoses transport gases from the gas regulators to a cutting torch. (1)

4.2.4 The flashback arrestors prevent flames from entering the gas hoses and gas cylinders. (2)

4.2.5 The working pressure gauge is set to a required pressure when cutting has to take place on a metal. (2)

4.3
- Undercutting
- Melting of metal
- Excessive slag
- Uneven cutting
- Overheating of metal (Any 2 × 1) (2)

4.4 4.4.1 A backfire is a loud noise caused by gases at the tip of the nozzle.

4.4.2 A flashback is a flame that burns back and enters the nozzle or torch.

(2 × 2) (4)

4.5 Causes of backfire

- Nozzle too close to the metal
- Foreign particles in the nozzle
- Incorrect gas flow
- Overheated nozzle
- Dirty nozzle (Any TWO) (2)

Causes of flashback

- Incorrect pressure
- Loose nozzle or adapter
- Failure to purge
- Kinked hose
- Blocked nozzle
- Overheating of nozzle or the torch (Any TWO) (2)

[25]

QUESTION 5

5.1 5.1.1 Down-hand welding position

5.1.2 2F

5.1.3 Vertical upward welding position

5.1.4 4F

5.1.5 6G

(5 × 1) (5)

5.2

- The gas cylinder valve and the air flow regulator valve are closed, and gas cannot flow.
- Hoses that are not connected correctly, that are bent, not properly secured or obstructed cause an insufficient gas flow.
- The tension on the feed rolls is not set correctly.
- The electrode liner is the incorrect size. As a result, the wire does not feed through loosely.
- The electrical cables are not connected correctly; as a result, there is no arc between the wire and the welding gun trigger.
- The setting up of the welding machine is incorrect, resulting in a poor weld. (Any 5 × 1) (5)

5.3

- Argon gas or shielding gas
- Carbon dioxide gas (2)

5.4

- Check the dimensions and alignment against the drawing specifications.
- Assemble the work piece according to the drawing specifications.
- Ensure that the work piece and area are safe.
- Check that the work piece is correct before welding commences. (4)

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5.5 5.5.1 Wire feeder feeds the welding wire at a specific and uniform rate to the work piece.

5.5.2 Power source is to supply direct current to the arc.

5.5.3 Welding gun has a switch which starts or stops the welding current, gas flow and the welding wire feed.

5.5.4 Wire spool supplies and holds the coiled welding wire.

(4 × 1) (4)
[20]

QUESTION 6

6.1 Open the acetylene valve on the torch and light up using a spark lighter. Open the valve more until the black smoke disappears and a carburising flame forms.✓ Open the oxygen valve on the torch until the flame turns almost purple in colour:✓ this is known as the oxidising flame.✓ The final step is to turn the oxygen valve on the torch until you get a well-defined white cone on the tip of the nozzle that is between 1.5 to 20 mm long.✓ This is known as a neutral flame.✓ (5)

6.2
- Ensure that the gas cylinders are upright and placed against a wall.
- Ensure that the gas cylinders are secured by using chains, a firm stand or brackets.
- Ensure that a collar or neck ring is placed over the cylinder valves.
- Always use a suitable trolley with chains, brackets or stands to move gas cylinders.
- Ensure that you do not use wire, rope or slings to hang or secure gas cylinder.
- Ensure that proper straps are used to transport cylinders from one point to another.

(Any 5 × 1) (5)
[10]

TOTAL: 100

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